

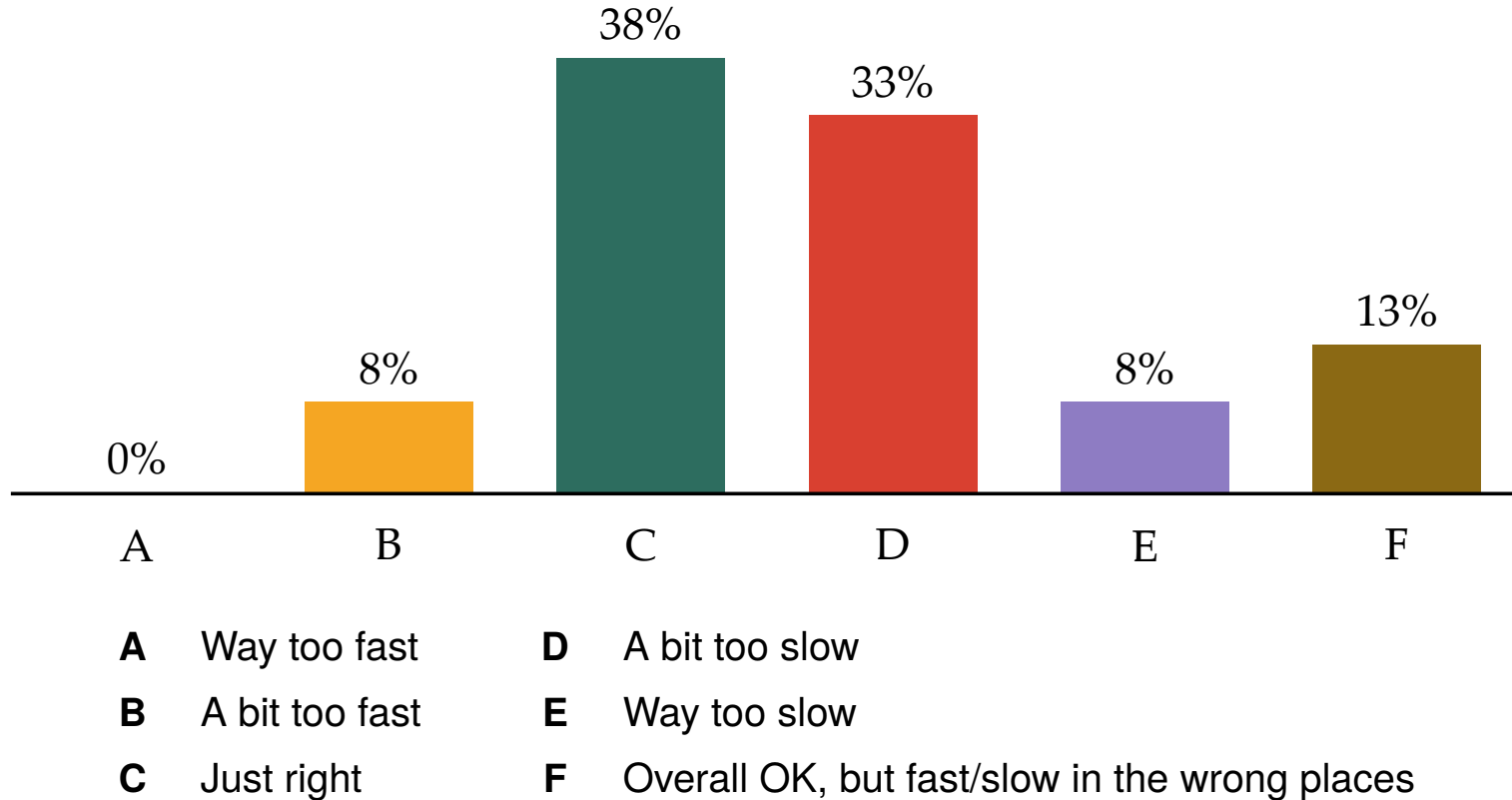
ECON 1550: International Finance

# Exchange Rates and the Foreign Exchange Market: An Asset Approach

# Announcements

- Problem Set 2 due Wednesday before midnight
- Survey result on speed of lecture
- Office hours and section locations

# Survey Results: Speed of Lecture



# Office Hours and Sections

| Day / Time        | Who          | Type                 | Location              |
|-------------------|--------------|----------------------|-----------------------|
| Mon 3:00–4:00 PM  | Prof. Duarte | Office hours         | Here!                 |
| Mon 4:00–6:00 PM  | Nathalie     | Section/Office hours | Maxcy Hall, Room 010  |
| Tue 4:00–5:00 PM  | Raisa        | Section              | TBD                   |
| Wed 12:00–2:00 PM | Eric         | Section/Office hours | Salomon 003           |
| Wed 3:00–3:45 PM  | Leo          | Office hours         | Sayles Hall Room 205  |
| Wed 4:00–5:00 PM  | Raisa        | Office hours         | Sayles Hall Room 205  |
| Wed 5:45–7:00 PM  | Leo          | Office hours         | TBD                   |
| Fri 12:00–1:00 PM | Prof. Duarte | Office hours         | 70 Waterman, Room 309 |

← CHECK !!

# Agenda

- Finish asset approach to exchange rate determination
  - UIP
  - Carry trade
- Next up: Money market

# A model of exchange rate determination

| Exogenous variables |                        | Endogenous variables |               |                               |                       |
|---------------------|------------------------|----------------------|---------------|-------------------------------|-----------------------|
| Variable            | Description            | Variable             | Description   | Equation                      | Type of equation      |
| $R$                 | Domestic interest rate | $E$                  | Exchange rate | $R = R^* + \frac{E^e - E}{E}$ | Equilibrium condition |
| $R^*$               | Foreign interest rate  |                      |               |                               |                       |
| $E^e$               | Expected exchange rate |                      |               |                               |                       |

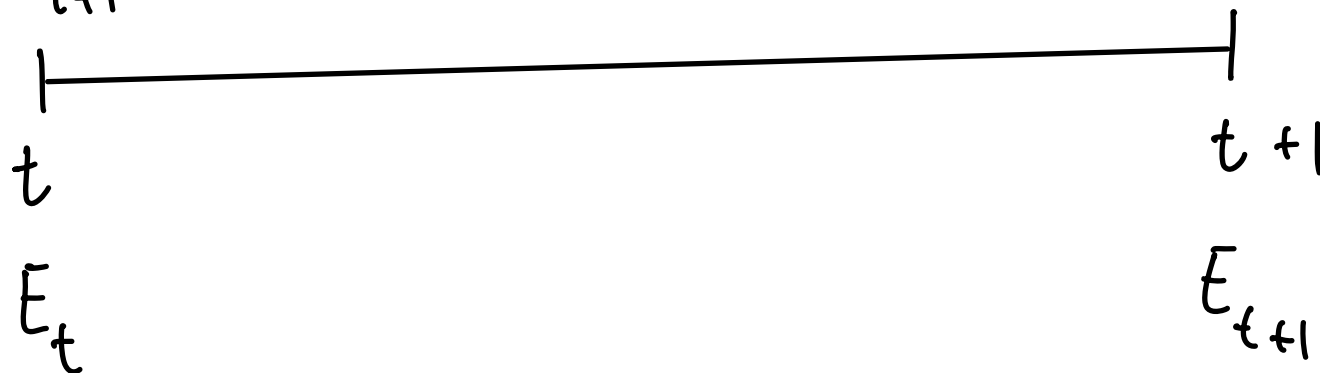
# A model of exchange rate determination

- Two investment opportunities
  - Domestic bond with return  $R_{\$}$  in Dollars
  - Foreign bond with return  $R^*$  in Euros
- To be indifferent between the two investments, the **uncovered interest parity condition** must hold:

$$(\text{UIP}): R_{\$} = R^* + \frac{E_{\$/\text{EUR}}^e}{E_{\$/\text{EUR}}} - 1$$

# Exchange Rates

$E_{t+1}^e$  BEST GUESS OF  $E_{t+1}$



$E_t$  IS # DOLLARS  
NEEDED TO BUY 1€  
AT  $t$

$E_t \uparrow$

DEPRECIATION OF \$

$E_{t+1}$  NOT  
KNOWN AT  
 $t$



# Bonds



$P_t$

$\$100$

$$R_t = \frac{\$100 - P_t}{P_t}$$

$P^*$

$R^*$

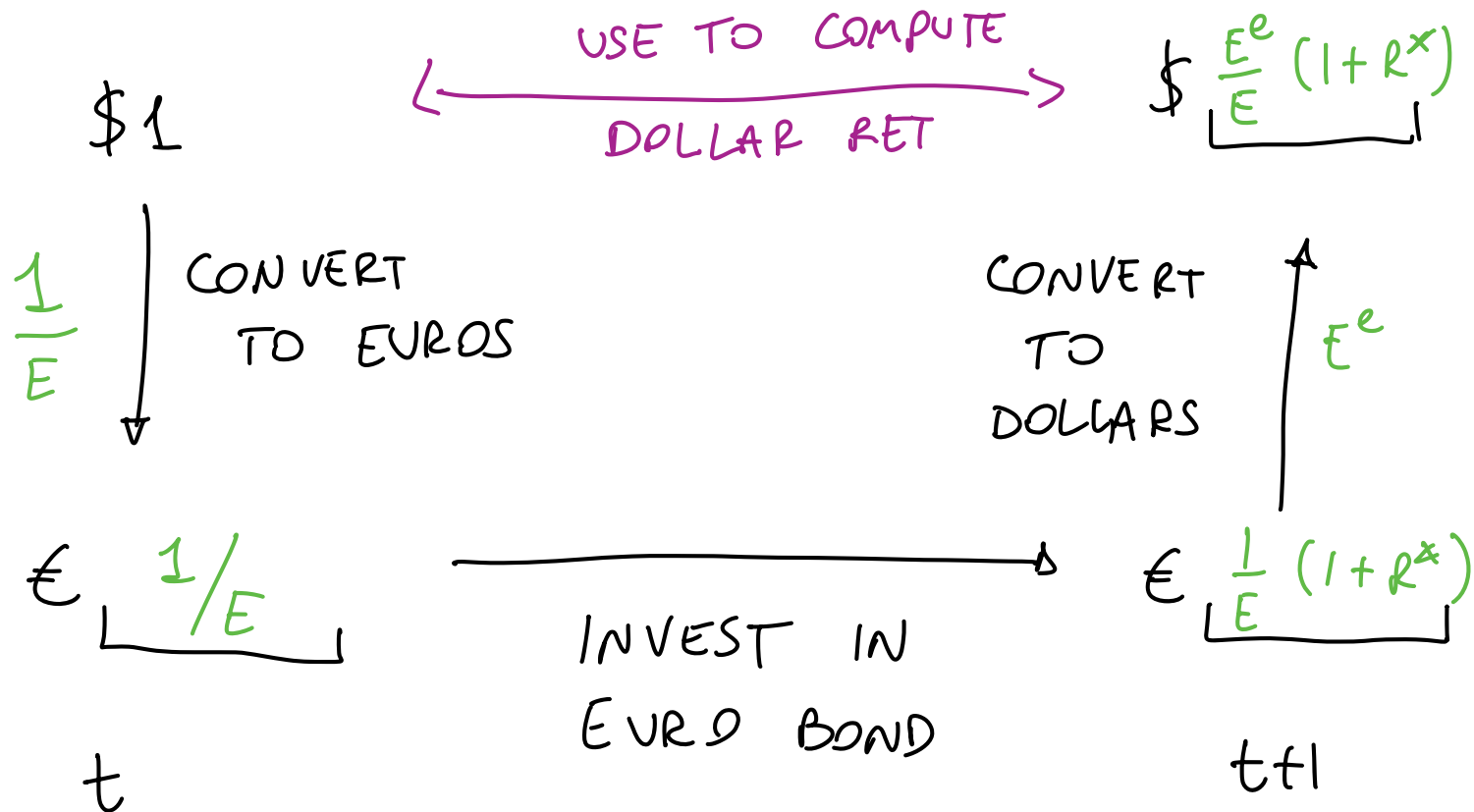
$\text{€}100$

# Realized vs Expected Returns

$$\text{DOLLAR BOND: } \underset{t}{\$1} \longrightarrow \underset{t+1}{\$1 \times (1+R)}$$

$$\text{EURO BOND: } \underset{t}{\text{€}1} \longrightarrow \underset{t+1}{\text{€}1 \times (1+R^*)}$$

# Expected Returns of Foreign Bond



# Approximations

$1 + R \rightarrow$  DOLLAR RET ON  
DOLLAR BOND

$\frac{E^e}{E} (1 + R^*) \rightarrow$  DOLLAR RET ON  
EURO BOND

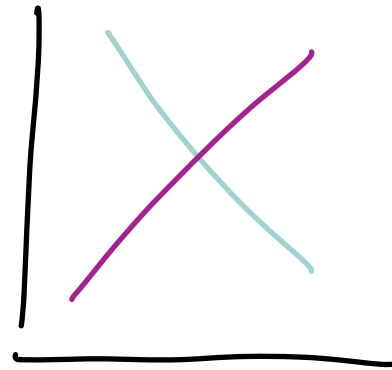
$$(UIP) \quad 1 + R = \frac{E^e}{E} (1 + R^*)$$

# Approximations

$$(1+x)(1+y) = 1 + x + y + \underbrace{xy}_{\text{IGNORE!}} \approx 1 + x + y$$

$$\frac{1+x}{1+y} \approx 1 + x - y$$

$$\frac{1+x}{1-y} \approx 1 + x + y$$



LINEAR!

# Uncovered Interest Parity

- For both strategies to have the same expected return

(EXACT UIP) :

$$1 + R_{\$} = \frac{E_{\$/\text{EUR}}^e}{E_{\$/\text{EUR}}} (1 + R^*)$$

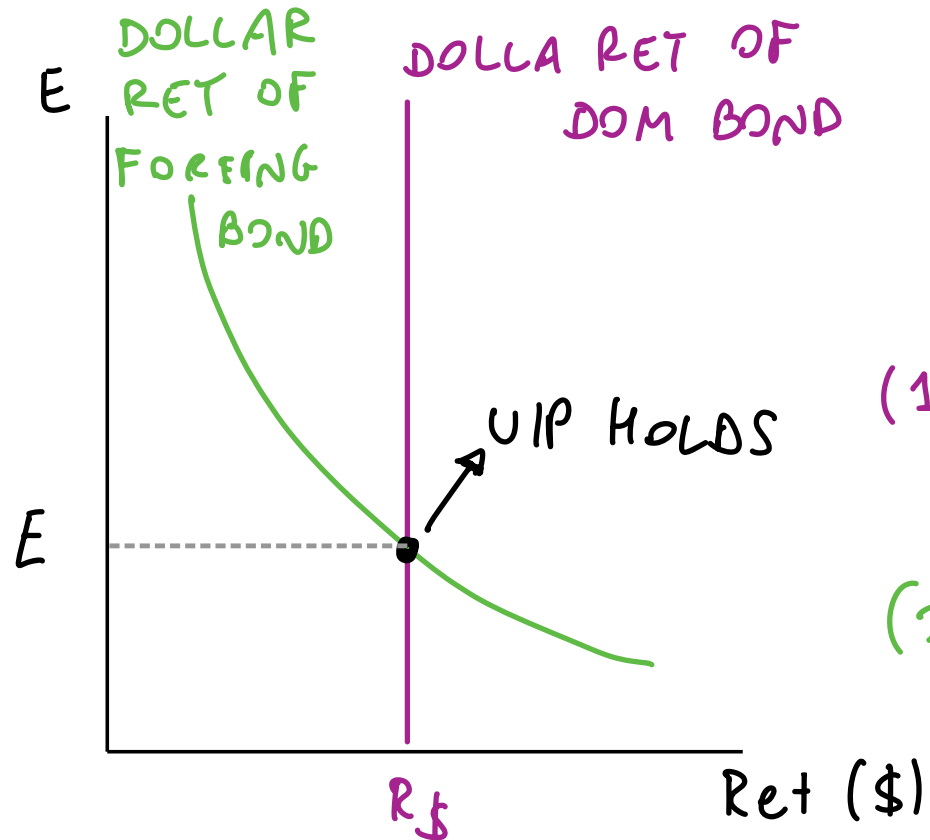
- Approximating

(UIP) :

$$R_{\$} = R^* + \frac{E_{\$/\text{EUR}}^e}{E_{\$/\text{EUR}}} - 1$$

$E \uparrow$  DEPR. DOLLAR WORTH LESS

# Equilibrium in Foreign Exchange Market

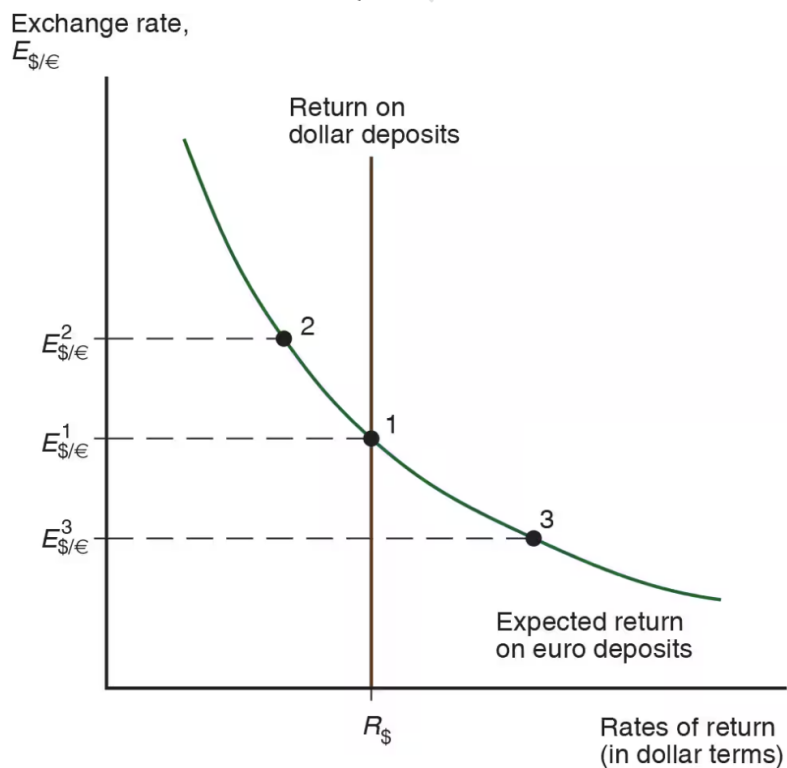


$$\text{UIP: } R_{\$} = R^* + \frac{E_{\$/\text{EUR}}^e}{E_{\$/\text{EUR}}} - 1$$

$$(1) R_{et} = R_{\$}$$

$$(2) R_{et} = R^* + \frac{E^e}{E} - 1$$

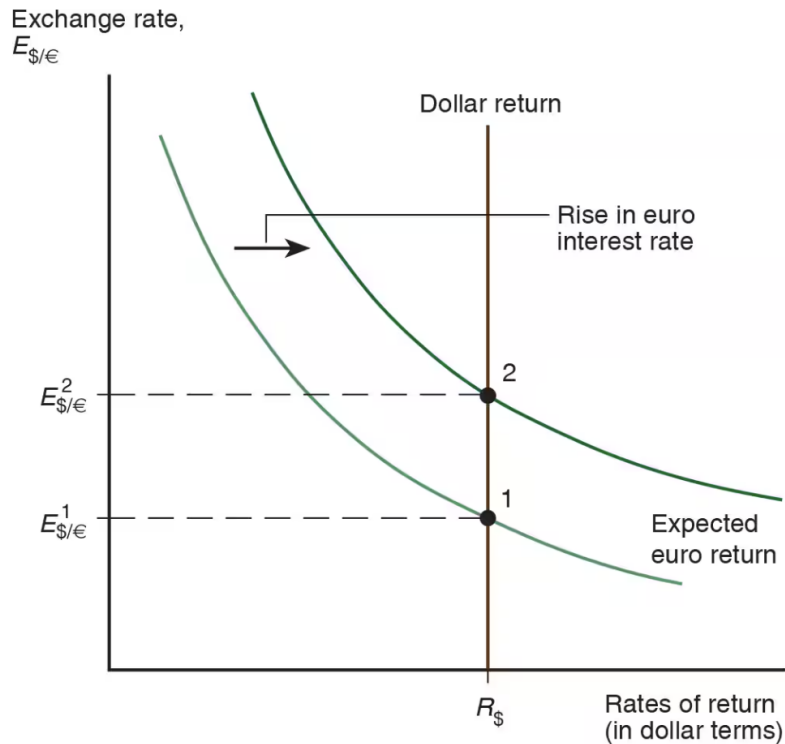
# Equilibrium in Foreign Exchange Market



$$\text{UIP: } R_{\$} = R^{*} + \frac{E_{\$/\text{EUR}}^e}{E_{\$/\text{EUR}}} - 1$$



# Shocks: Rise in Euro Interest Rate



$$R^* \uparrow$$

$$(1) R_{\text{eff}} = R_{\$}$$

$$(2) R_{\text{eff}} = R^* + \frac{E^e}{E} - 1$$

SHIFT OF (2) TO  
THE RIGHT.

$E \uparrow$  \$ DEPRECIATES!

# The carry trade

- Borrowing at “low” rate  $R$  and lending at “high” rate  $R^*$  is a **carry trade**

$$\begin{array}{l} \text{expected return} \\ \text{on carry trade} \end{array} = R^* + \left( \frac{E^e}{E} - 1 \right) - R + \text{risk premium}$$

- Risk: Future exchange rate is not known when we start the carry trade,  $E^e$  can be different from the realized future exchange rate

# Empirical Failure of UIP

Eight currency portfolios sorted by interest rate differential (portfolio 1 = lowest rates, portfolio 8 = highest).

High interest rate currencies earn higher mean excess returns and have higher Sharpe ratios.

Annual data, 1953–2002, US investor perspective.

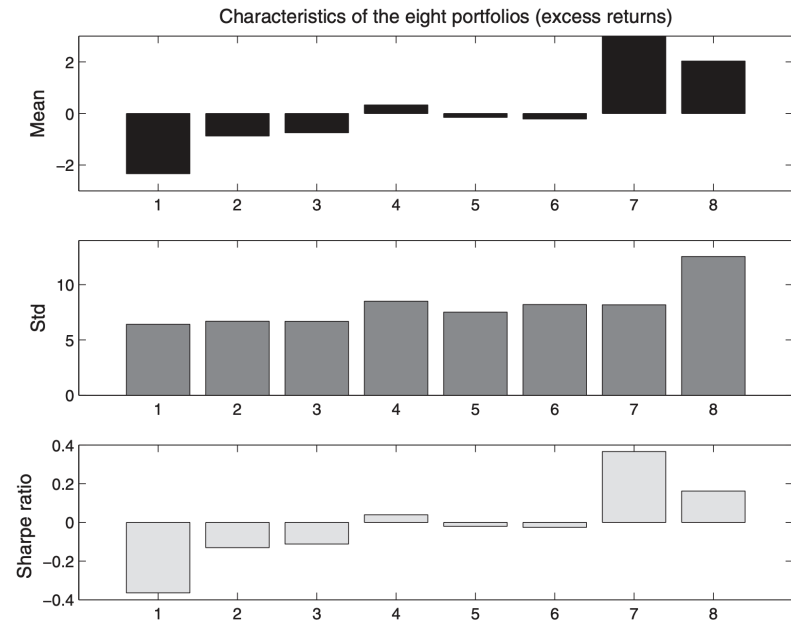


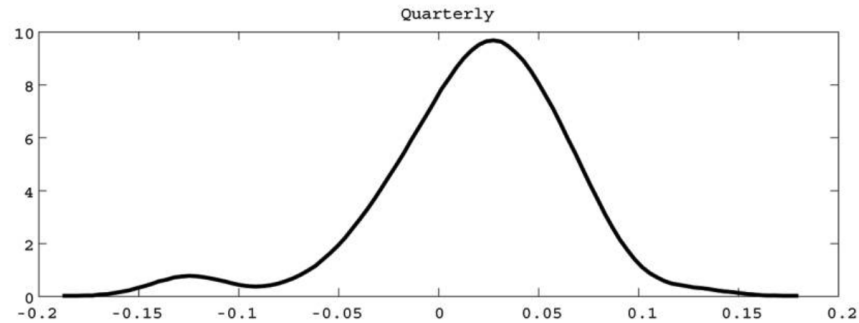
FIGURE 1. EIGHT CURRENCY PORTFOLIOS

Source: Lustig & Verdelhan (2007), *American Economic Review*

# Explanations For Carry Risk Premium

- Crash risk / peso problem
- Correlation with consumption
- Global Volatility Risk
- Liquidity
- Constrained intermediaries

# Crash Risk



Kernel density of excess returns on a carry trade portfolio (long three high interest currencies, short three low interest currencies). Right: USD/JPY exchange rate, 1996–2000.



Fig. 1. U.S. dollar/Japanese yen exchange rate from 1996 to 2000

Source: Brunnermeier, Nagel, and Pedersen (2008), *NBER Macroeconomics Annual*

# Consumption Risk

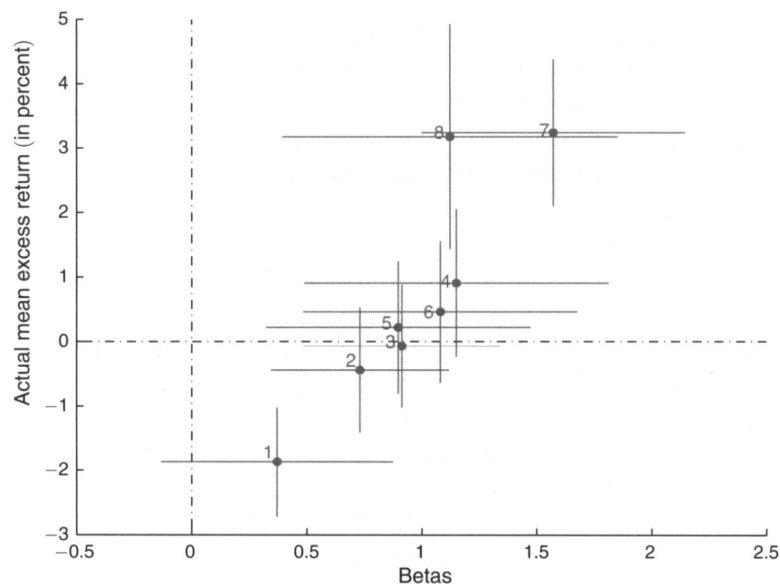
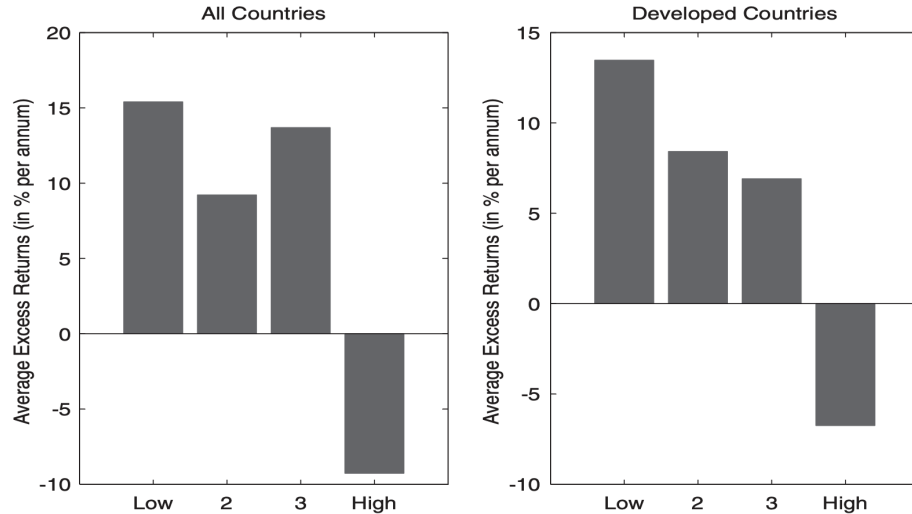


FIGURE 3. DURABLE CONSUMPTION GROWTH BETAS AND AVERAGE CURRENCY EXCESS RETURNS

The dots represent point estimates; the lines represent one standard deviation above and below. The sample is 1953–2008, annual data. Higher consumption betas correspond to higher currency excess returns.

Source: Lustig & Verdelhan (2011), *American Economic Review*

# Global Volatility Risk

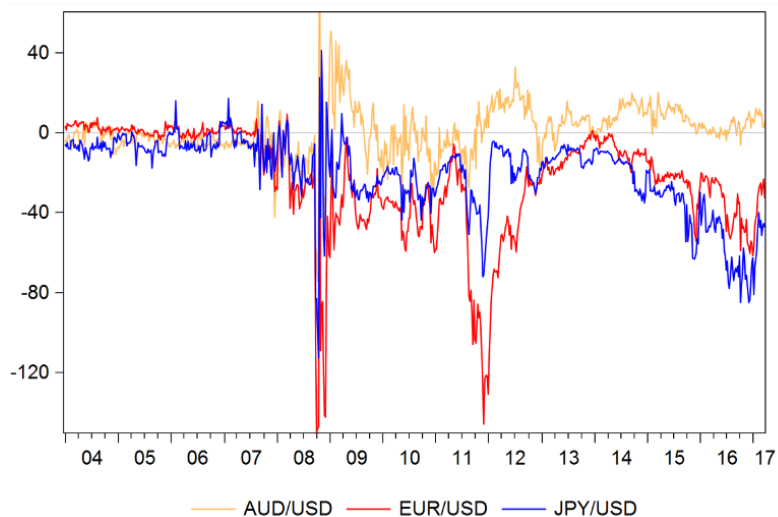


**Figure 2. Excess returns and volatility.** The figure shows mean excess returns for carry trade portfolios conditional on global FX volatility innovations being within the lowest to highest quartile of its sample distribution (four categories from “Low” to “High” shown on the x-axis of each panel). The bars show average excess returns for being long in portfolio 5 (largest forward discounts) and short in portfolio 1 (lowest forward discounts). The left panel shows results for all countries, while the right panel shows results for developed countries. The sample period is November 1983 to August 2009.

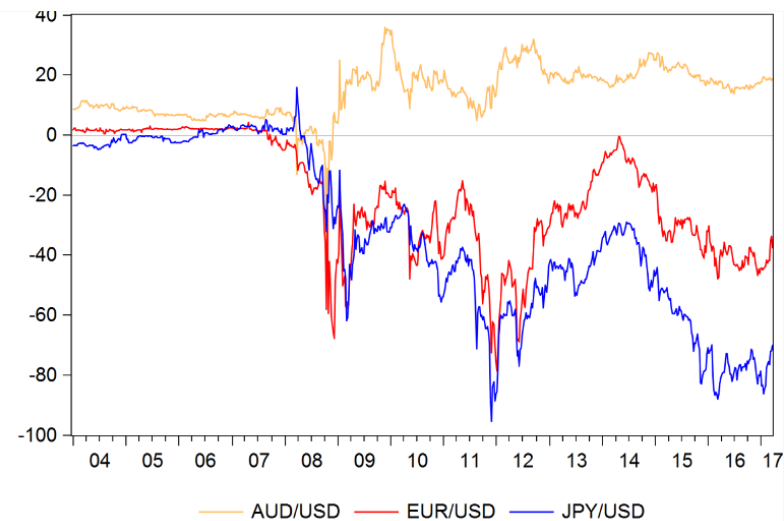
Mean excess returns for carry trade portfolios conditional on global FX volatility innovations being within the lowest to highest quartile. Carry trade earns high returns when volatility is low, but suffers large losses when volatility is high.

Source: Menkhoff, Sarno, Schmeling, and Schrimpf (2012), *Journal of Finance*

# Violations of *Covered* Interest Parity



3-month basis:  $b_{3m}$



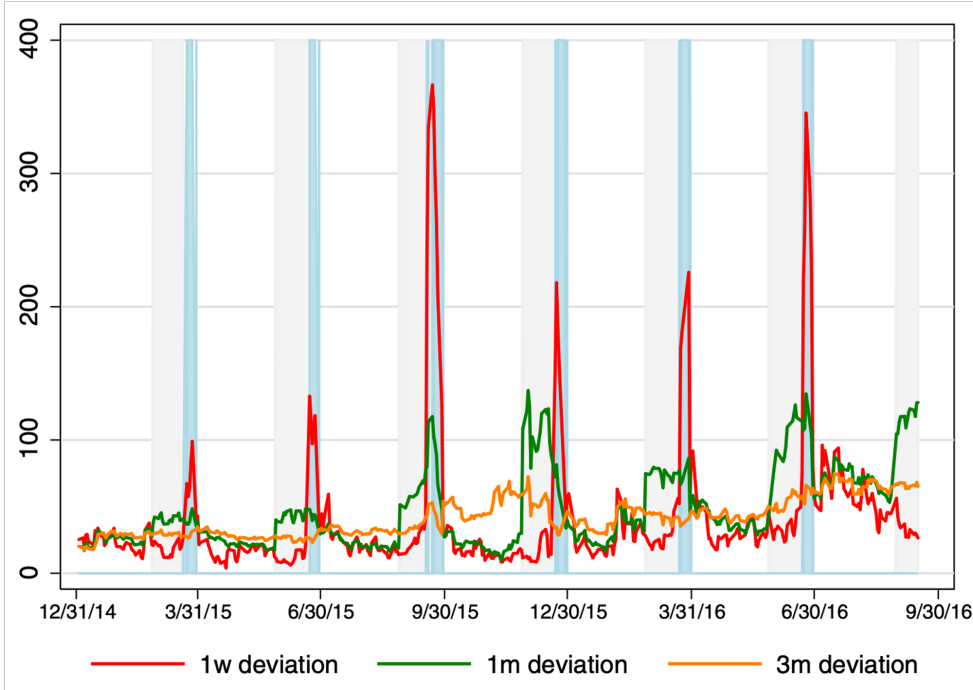
3-year basis:  $b_{3y}$

Source: [Sushko, Borio, McCauley, and McGuire, "The Failure of Covered Interest Parity"](#)



# Intermediaries are Constrained

Illustration of quarter-end dynamics of CIP deviations.



Source: Du, Tepper, and Verdelhan, “Deviations from Covered Interest Rate Parity,” *Journal of Finance*

